

Coordination vs. Competition: Local Government Centralization and Metropolitan Growth

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1 Introduction

A longstanding question in economics asks how government should be organized. Because public goods lack typical market-based provision mechanisms (Musgrave (1939); Samuelson (1954)), economists have long worried about the efficiency of their supply. A foundational contribution by Oates (1972) stresses that the organization of government itself shapes these outcomes, framing the central debate: should public services be centralized into a few large jurisdictions, or decentralized across many smaller ones? While the debate occurs across all levels of government, it is particularly salient at the local level. In this paper, I present empirical evidence on the effect of centralization on metropolitan growth using variation across the metropolitan statistical areas (MSA) of the United States. Using an IV strategy based on the Postal Service’s introduction of Rural Free Delivery in 1902, I find that centralization causes both higher population and employment growth between the late 1970’s and the late 2010’s. I also find that centralization causes higher local government employment, a greater amount of revenue to be collected, and fewer data centers to locate within the MSA.

A key tension in the debate over centralization arises over the role of competition between governments. The decentralized position favors competition and focuses on frameworks featuring information and incentives inspired by Tiebout (1956) and Hayek (1945). In contrast, the centralized position favors coordination and worries that externalities can sour decentralized decision-making, formalized in the tax competition models of Wilson (1986), Zodrow and Mieszkowski (1986), and Wildasin (1988).

The questions surrounding centralization have direct policy relevance, not just academic interest. Many countries have actively restructured local governments in recent decades, often through mergers that increase centralization. Between 2008 and 2017, fifteen European countries implemented large-scale municipal reforms (Swianiewicz (2018)). Japan undertook a similar consolidation in the early 2000s, reducing its number of municipalities from 3,232 to 1,719 (Miyazaki (2014)). Toronto merged its six core municipalities into a single government unit in 1999 (Slack and Bird (2013)). In the United States, voters have repeatedly considered local government consolidation, including 166 city–county merger referenda since 1815 (Hall, Matti, and Zhou (2020)), alongside numerous smaller merger and annexation decisions. Because these reforms represent discrete increases in centralization, evaluating their effects or conducting ex-ante counterfactual analysis depends critically on the behavioral framework used to characterize local government interactions.

Two main empirical challenges obscure the debate. First, the effect of centralization on many local government outcomes can be consistent with either framework. For example, the higher revenue collection that I find in more centralized MSAs can be rationalized within the competition framework through weaker competitive pressure across jurisdictions, allowing governments to extract greater revenue from residents (Brennan and Buchanan (1980)), or within the coordination framework through the internalization of fiscal externalities. In tax competition models, local tax reaction functions are typically estimated as strategic complements, implying that tax reductions by one jurisdiction induce reductions by neighboring jurisdictions and generate inefficiently low levels of public good provision when these spillovers are not internalized through centralization (Brueckner (2003)).

However, a comparative static over which the two frameworks disagree concerns growth. In spatial equilibrium models (Rosen (1979); Roback (1982); Moretti (2012); Redding and Rossi-Hansberg (2017)), growth reflects differences in expected utility, with locations offering higher expected utility experiencing greater growth. Interpreting growth through this revealed-preference lens allows differences in growth to serve as a measure of relative economic performance. The competitive framework predicts that greater centralization reduces growth through the removal of market-like discipline, while the coordination framework predicts that greater centralization increases growth through the internalization of externalities. These opposing predictions define the central empirical application of this paper.

The second empirical challenge concerns endogeneity. The organization of local government is in part shaped by a tradeoff between preference heterogeneity over public good bundles pushing towards decentralization, and technology affecting economies of scale pushing towards centralization (Alesina and Spolaore (1997); Alesina, Baqir, and Hoxby (2004)).

Because these underlying forces are unobserved and may also directly affect metropolitan growth and the other local government outcomes I study, OLS estimates will generally be biased because they confound the causal effect of centralization with these omitted determinants.

To address the endogeneity problem, I use an IV strategy based on the Postal Service’s implementation of Rural Free Delivery (RFD) in 1902. Using a complete census of post offices from Blevins and Helbock (2021), I instrument centralization with the number of post office closures in each MSA between 1902 and 1910. Prior to RFD, what were considered rural residents at the turn of the century typically collected mail at local post offices, whose viability depended on the postal activity generated by surrounding communities. As an extension of the 1863 Free City Delivery program to rural areas (United States Postal Service (2024)), RFD shifted away from local pickup to direct household service, reducing the need for many small and geographically dispersed post offices. The resulting closures were disproportionately concentrated in rural areas and therefore provide a proxy for the historical dispersion of settlement.

The RFD post office closure instrument is highly predictive of modern local government decentralization. I interpret the instrument as operating through the municipal annexation process, as many jurisdictions incorporate as a legal defense against annexation attempts by established municipalities (Burns (1994)). Because a greater number of post office closures proxies for a larger number of historically dispersed and unincorporated settlements, it increases the number of potential annexation margins within an MSA. As the number of annexation opportunities rises, so does the number of opportunities for defensive incorporation, producing a more decentralized local government structure.

The identifying assumption is that the historical settlement dispersion proxied for by the number of post office closures affects the modern outcomes I study only through its effect on the organization of local government. A potential concern is that the decision to accept or resist annexation may itself reflect the same unobserved determinants that generate the modern endogeneity problem. However, the underlying determinants of both preference heterogeneity and economies of scale are highly time-varying. Technological changes in transportation, communication, and public service provision substantially alter the efficient scale of governance, while preferences over local public goods also evolve over time. In contrast, political boundaries established at a point in time are highly durable and likely to persist despite changes in the underlying economic environment.

To measure modern centralization, I use data from the Annual Survey of Public Employment and Payroll (ASPEP), which reports employment in each government unit for 1987, 1997,

2007, and 2017. Because centralization is not directly observed but rather is a latent concept, I construct an index based on three observed measures: the share of MSA local government employment in the largest local government, the number of governments, and a Herfindahl-Hirschman index of local government employment shares. Each measure is reflective of latent centralization, allowing me to combine their joint variation across MSAs and over time to recover a common component that I interpret as the underlying level of centralization. Importantly, the results I find remain qualitatively consistent regardless of the aggregation scheme used to create the index, or if I use the observed measures alone.

Using this strategy, I find that greater centralization causes higher levels of growth. I measure growth in two ways. First, I use SEER data from the CDC to measure MSA population levels every decade starting in 1977 and ending in 2017, with years selected to coincide with the ASPEP data. Second, I measure private employment across the same years using the Business Dynamics Statistics dataset. I calculate growth using each year as a base and 2017 as the terminal period. Across both measures, and for all but one population growth horizon, greater centralization is found to increase growth, providing evidence in favor of the coordination framework.

This relationship is consistent with a basic feature of local government: jurisdictions interact over shared tax bases, mobile residents, and spatially diffuse public goods (Sinn (1997)). In decentralized settings, governments may not fully internalize the fiscal externalities generated by their policy choices. Competition across jurisdictions can therefore suppress equilibrium tax rates and reduce public good provision. By internalizing these spillovers, centralization can mitigate this coordination failure and increase growth.

However, while the growth results are consistent with coordination failure, they do not rule out alternative channels such as economies of scale in public good provision. The remaining empirical exercises therefore examine whether the data are more consistent with a coordination failure interpretation or a scale-based efficiency channel.

I confront two primary challenges in testing for economies of scale. First, I do not observe public good output directly. Second, public goods are inherently multi-dimensional bundles. Even if all dimensions were observable, differences in observed provision could reflect tradeoffs involving unobserved preferences rather than production efficiency. For example, higher quality roads in one MSA may reflect a stronger preference for roads over parks rather than more efficient provision. To partially address this issue, I examine local government employment as a measure of input use. I find that more centralized MSAs employ more labor, which is inconsistent with a simple labor-saving economies of scale channel.

Next, I find that more centralized local governments collect both more total revenue and

more tax revenue conditional on population. This pattern directly matches the predictions of tax competition models. In decentralized MSAs, taxing authority is fragmented across jurisdictions, intensifying competition for a mobile tax base. Strategic complementarities in local tax reaction functions can suppress equilibrium tax rates and aggregate revenue when fiscal spillovers are not internalized, allowing more centralized MSAs to collect greater revenue through the internalization of these externalities.

The joint results of higher labor input use and revenue collection in centralized MSAs further weaken a pure economies of scale interpretation. Greater economies of scale should allow governments to produce a given level of public goods at lower cost, or a greater level of public goods at constant cost. Instead, centralized MSAs appear to expand both inputs and revenue collection beyond what a pure economies of scale mechanism would predict. A natural interpretation is that decentralized governments face constraints on public good provision, and that centralization alleviates these constraints in a manner consistent with the coordination failure mechanism.

In a final empirical exercise, I examine location decisions made by firms constructing data centers. Data centers are relatively new, highly mobile, capital-intensive, and particularly sensitive to local tax policy. These attributes make them a close empirical analogue of the mobile capital in tax competition models. As a useful analogy, a firm's location decision resembles a seller's choice of which auction to enter. More decentralized MSAs, with a larger number of local governments, resemble settings with more competing bidders. If coordination failures lead to lower effective tax rates or regulatory burdens, firms have an incentive to disproportionately locate in decentralized MSAs where governments compete more aggressively for investment, similar to a seller's incentive to enter an auction with more bidders. Consistent with this mechanism, I find that decentralized MSAs are more likely to host at least one data center, suggesting that coordination failures are sufficiently strong to affect firm location decisions.

Taken together, the positive relationship between centralization and growth suggests that models of coordination failure provide a more accurate description of local government behavior than models of local government competition. While not conclusive, the evidence further suggests that the positive effect of centralization on growth is driven primarily by the alleviation of coordination failures rather than by economies of scale. Because interactions among local governments are inherently spatial, spillovers and externalities are particularly salient within MSAs. These externalities can distort competition across decentralized jurisdictions and inefficiently suppress public good provision. By internalizing these spillovers, centralized local governments can mitigate coordination failures and increase the provision

of public goods, contributing to higher levels of growth.

This work contributes to several literatures, beginning with research on political mergers. Because political mergers generate discrete increases in centralization, estimating the causal effect of centralization is central to evaluating these reforms. Existing studies, largely based on case studies and quasi-experimental designs, examine a wide range of institutional contexts, including Israel (Reingewertz (2012)), China (Tang and Hewings (2017)), Denmark (Blom-Hansen et al. (2016)), Japan (Miyazaki (2018)), and Iceland (Karlsson and Eythórs-son (2022)). Recent syntheses emphasize the highly heterogeneous and often contradictory results across settings and outcomes (Zheng, Sun, and De Witte (2023)). In contrast, I estimate the causal effect of centralization across relatively homogeneous U.S. metropolitan areas using variation generated by a historical policy shock. By abstracting from discrete merger events, this approach mitigates concerns about institutional heterogeneity and endogenous selection that complicate merger-based analyses.

This paper also relates to the broader literature on government centralization and institutional performance. A large macro-level literature studies how cross-country variation in fiscal and political centralization between national and sub-national governments affects growth and institutional outcomes (Barro (1991); Weingast (1995); Sala-i-Martin (1997); Davoodi and Zou (1998); Treisman (2000); Rodden (2004); Iimi (2005); Baskaran and Feld (2013)). In contrast to my findings, this literature often concludes that decentralization promotes economic growth (Dillinger (1994)). One interpretation is that the relevant trade-offs differ by scale. At the national level, decentralization may intensify competition across large and geographically separated jurisdictions where spillovers are less salient. At the metropolitan level, however, the dominant concern may instead be the internalization of spillovers among closely linked governments, making greater centralization more effective.

Finally, this paper fits most closely within a strand of literature examining within-country variation in local governmental structure and centralization. Examples include Hoxby (2000) on educational performance, Stansel (2005) on population growth, Hatfield and Kosec (2013) and Hatfield and Kosec (2019) on wage growth and pollution, Hatfield, Kosec, and Rodgers (2024) on housing prices, Hobbs, Millsap, and Stansel (2019) on economic development, and Bianchi, Giorcelli, and Martino (2023) on labor market outcomes. However, existing studies rely on a wide range of centralization measures and sources of identifying variation that do not map cleanly to local government centralization as measured here. Differences in how centralization is defined and identified may therefore help explain the divergence between prior findings and the results of this paper.

The remainder of the paper proceeds as follows. Section 2 presents the data on local govern-

ment organization, including the measurement of centralization and the outcomes I study. Section 3 outlines the empirical strategy and describes the instrumental variables used for identification. Section 4 reports the results and discusses their implications for understanding the mechanisms of centralization. Section 5 concludes.

2 Data

The main unit of analysis in this study is the metropolitan statistical area (MSA), as defined by the Office of Management and Budget. An MSA consists of a group of counties organized around an urban core with substantial economic integration, as measured by commuting flows. This definition provides a useful approximation to the relevant market for local government interaction. In tax competition models, equilibrium requires that each government's policy is a best response to the policies of other jurisdictions. Concretely, I treat the MSA as the primary strategic environment in which local governments respond to one another's policies.

The median MSA contains two counties, while 159 of the 378 MSAs consist of a single county. The Atlanta–Sand Springs–Roswell MSA contains the most counties with 29. Because commuting patterns and urbanization evolve over time, MSA boundaries change across decades. To maintain a consistent geographic unit, I fix MSA boundaries at their 2017 definitions and use these boundaries to harmonize all historical variables. This allows changes over time to be interpreted within a consistent geographic unit.

I measure local government organization using the 1987, 1997, 2007, and 2017 waves of the Annual Survey of Public Employment and Payroll (ASPEP). The ASPEP is a full census of local governments conducted every five years and reports employment for each individual government unit. I apply several restrictions to the data. I drop all government units with zero employment. I exclude school districts, as they represent a distinct margin of governance, financing, and labor-market organization. To maintain consistency, I also drop all education employment from other government units, as well as all hospital employment. I then match each government unit to its corresponding MSA.

2.1 Local Government Centralization Index

I construct five centralization indices: one for each survey year and one using the joint variation across all years. The indices are constructed by first deriving proxy measures for

centralization from the ASPEP employment data, residualizing the measures on population, land area, and surface water area, then aggregating the residualized measures into an index.

MSA m in survey year t can be described by the vector E_{mt} , whose entries correspond to employment in individual government units. From these vectors, I construct three proxy measures for centralization for each MSA. The first is the number of governments with nonzero employment. Holding population constant, an MSA with many governments is considered decentralized, while one with few governments is considered centralized. The second is the employment share of the largest government unit, analogous to a one-firm concentration ratio. As this measure rises, employment becomes more concentrated in the largest unit, indicating higher centralization. The third is a Herfindahl–Hirschman index (HHI) based on government employment shares, commonly used in the industrial organization literature. A fully centralized MSA has an HHI of one, while a fully decentralized MSA approaches zero.

The three measures are given by:

$$\log |E_{mt}|, \quad \frac{\max_m E_{mt}}{\sum_k E_{kt}}, \quad \sum_{m=1}^{|E_{mt}|} \left(\frac{E_{mt}}{\sum_k E_{kt}} \right)^2$$

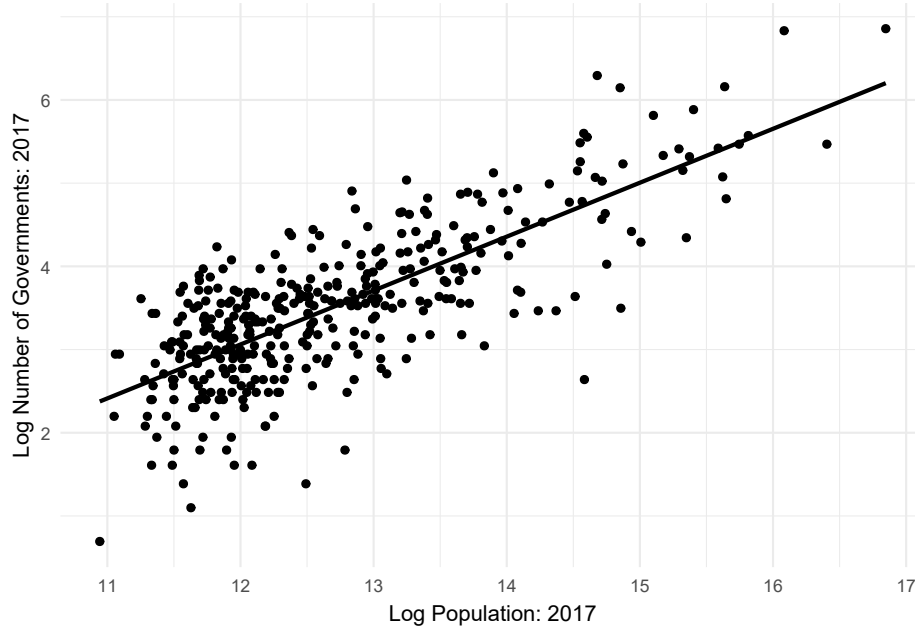
These measures can be linked through a diversity index popular in the biology literature (Hill (1973); Maturo (2018)) and the industrial organization literature on concentration (Hannah and Kay (1977); Nocke and Schutz (2025)). The three measures I use are specific cases of a family of measures given by

$$H(q) = \left(\sum_{i=1}^{|E_{mt}|} s_{im}^q \right)^{\frac{1}{1-q}}, \quad s_{im} = \frac{E_{imt}}{\sum_k E_{kmt}}$$

Using this notation, $H(0)$ gives the number of governments measure, $\lim_{q \rightarrow \infty} H(q)$ approaches the inverse of the employment share for the largest government unit, and $H(2)$ gives the inverse of the HHI measure. The measures I use are monotone transformations of focal points within this family.

Because the interpretation of these measures depends on MSA scale and geography, I residualize each proxy on log MSA population in year t , log land area, and the log area of surface water. This residualization step is crucial for the interpretation of centralization. Figure 2.1 highlights this dependence. The horizontal axis plots log MSA population in 2017 and the vertical axis plots the log number of governments in 2017.

Figure 2.1: Number of Governments and Population



Notes: The figure plots the log number of local governments against log MSA population in 2017 for the 378 MSAs in the sample. Each point is an MSA and the solid line is the OLS fit.

Because the number of governments scales with population size, a fixed number of governments can represent different levels of centralization. For example, 100 local governments in a small MSA likely implies decentralization, while 100 governments in a large MSA likely indicates centralization. This intuition can be seen in Figure 2.1. Vertical variation between MSAs at the same population corresponds to changes in centralization, while horizontal variation does not. In addition, land area and in particular surface water area may constitute geographic differences across MSAs that affect the interpretation of centralization. By residualizing the measures on these factors, I isolate variation in organizational structure beyond scale and geographic effects.

I then apply principal component analysis to the residualized proxies and use the first principal component as the centralization index. I construct indices separately for each year using the three measures and a pooled index using all years and measures. The first principal component explains 77% of the variation in the single-year indices on average and 73% in the pooled index, indicating that the index captures a substantial source of common variation across the proxy measures.

Table 1 examines persistence in centralization over time. The 2017 centralization index is regressed on the indices from 2007, 1997, and 1987 in columns (1)-(3) respectively. Perfect persistence would correspond to a coefficient of one.

Table 2.1: Persistence in Local Government Centralization

	<i>Dependent variable:</i>		
	Centralization Index: 2017		
	(1)	(2)	(3)
Centralization Index: 2007	0.964*** (0.014)		
Centralization Index: 1997		0.927*** (0.019)	
Centralization Index: 1987			0.880*** (0.025)
Observations	378	378	378
R ²	0.929	0.859	0.774

*p<0.1; **p<0.05; ***p<0.01

Notes: Each column reports an OLS regression of the standardized 2017 centralization index on the standardized centralization index from the year listed in the row label. The sample includes 378 MSAs. Standard errors are reported in parentheses.

The results indicate a high degree of persistence: MSAs that are more decentralized in earlier periods tend to remain decentralized in later periods. I also construct alternative indices using a simple average of the residualized proxies and a factor model. These measures are highly correlated with the PCA-based index, and the main results are robust to their use, or the direct use of the observed measures.

2.2 Outcomes

2.2.1 MSA Population and Employment Growth

I construct MSA-level population using county-level data from the CDC SEER database, aggregated to 2017 MSA definitions. Population is measured for 1977, 1987, 1997, 2007, and 2017. I define growth as the log difference between population in 2017 and population in a base year. In the shortest horizon (2007–2017), average MSA population growth is 7.3%, while over the longest horizon (1977–2017) it is 55.2%.

I construct employment measures using the Business Dynamics Statistics (BDS) dataset. Because these data begin in 1978, I use employment levels for 1978, 1987, 1997, 2007, and 2017. Employment growth is measured analogously using log differences relative to 2017.

Because the ASPEP data for 1977 is unavailable, the estimates of centralization on growth between 1977 and 2017 rely on extrapolation. Table 2.1 shows that centralization is highly

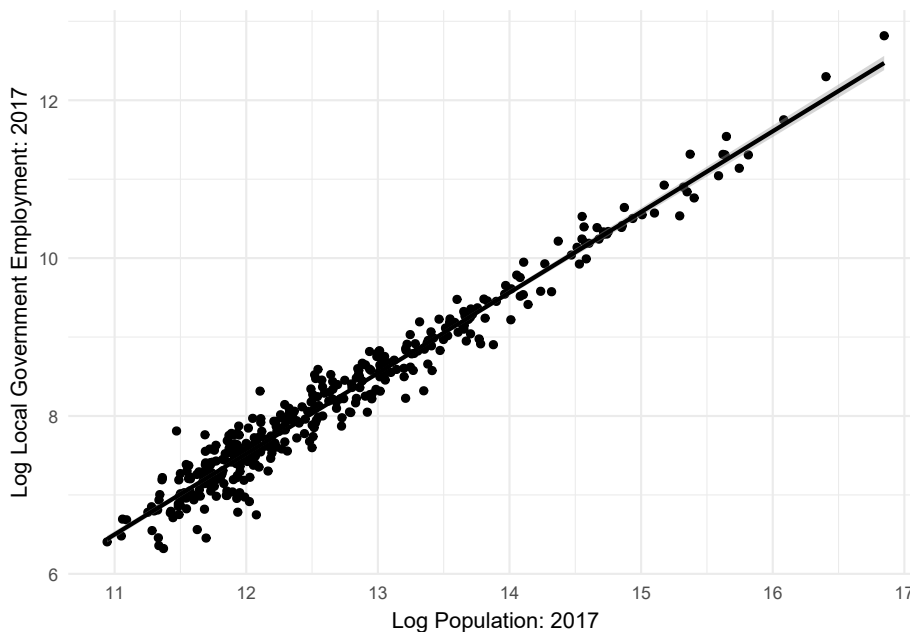
persistent across time, leading me to have high confidence when using the measure of centralization in 1987 to capture the level of centralization in 1977.

2.2.2 Local Government Employment and Finances

Local government employment is measured using the same ASPEP data used to construct the centralization index. Total employment is defined as the sum of employment across all government units within an MSA.

Figure 2.2 plots log local government employment against log population for 2017.

Figure 2.2: Local Government Employment and Population



Notes: The figure plots log local government employment against log MSA population in 2017 for the 378 MSAs in the sample. Each point is an MSA, the solid line is the OLS fit, and the shaded region is the 95% confidence interval.

The strong log-linear relationship motivates the log-log empirical specification used in the analysis.

To study government finances, I use data from the Census of Governments (CoG), which complements the ASPEP by providing financial information for each government unit. I extract total revenue and tax revenue for each government unit. I maintain the same sample restrictions as above, but I do not drop units with zero employment. This ensures that revenue from jurisdictions such as single-purpose taxing districts is included.

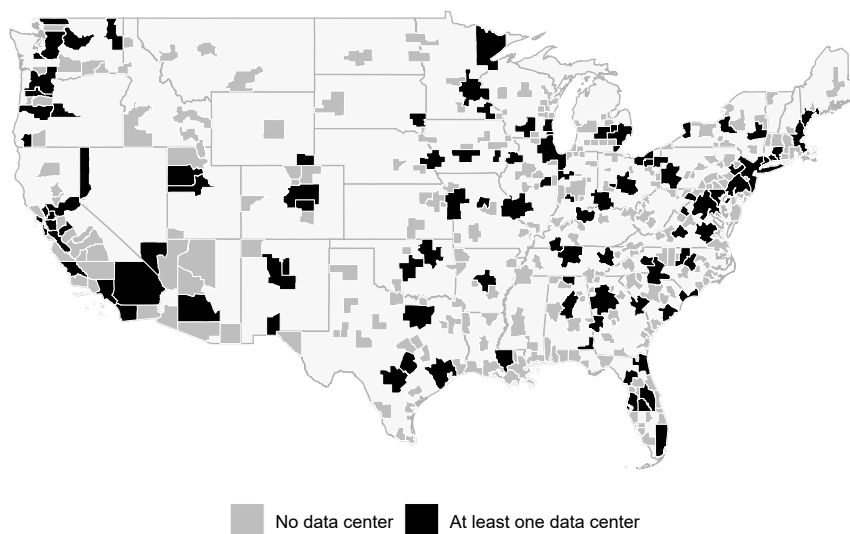
A major difference between the local government employment data and the financial data is the unit of measurement. Local government employment is measured in number of workers while finances are measured in dollars. This presents a measurement issue as the value of a dollar varies significantly across geographies (Moretti (2013)). To account for differences in local price levels, I use income data from the BEA along with the employment data from the BDS. The ratio of total income and total employment in survey year t captures average wage levels and helps account for differences in the total revenue and tax revenue between MSAs that are not driven by differences in organization.

2.2.3 Data Centers

The final outcome is the location of data center facilities. I use data from the IM3 Open Source Data Center Atlas, which reports 1,242 data center facilities and their county locations. I aggregate these data to the MSA level and construct an indicator for whether an MSA hosts at least one data center. Approximately 27% of MSAs have at least one facility.

Figure 2.3 displays the geographic distribution of data centers across MSAs. MSAs shaded in black host at least one data center, while those in grey do not.

Figure 2.3: Data Center Facility Locations



Notes: The map reports whether an MSA hosts at least one data center facility. MSAs shaded in black host at least one facility and MSAs shaded in gray do not. Data center locations are aggregated to 2017 MSA definitions.

2.2.4 Time Invariant Covariates

At its base, the empirical strategy that I discuss in the next section regresses the outcomes above on the centralization index. In these regressions, I include a set of time invariant covariates. These include land and surface water area mentioned above, an indicator for whether the largest government in the MSA is a county or a municipality, and an indicator for the presence of a port in the MSA. Additionally, I include the number of years between the creation of the first post office in the MSA and 1902, which is important for controlling for aspects of the instrument detailed below.

3 Empirical Strategy

This section presents the empirical specifications used to estimate the effect of centralization on the outcomes discussed above. The first subsection introduces the estimating equations, while the second details the instrumental variables strategy and associated first-stage robustness checks.

3.1 Empirical Specifications

This subsection details the empirical specifications I use to estimate the effect of centralization on population and employment growth, local government employment levels, local government finances, and the location of data centers. The value of the centralization index in MSA m from survey year t is denoted as C_{mt} . In the growth equations, I use the centralization index constructed using the joint variation across years and drop the t index. Throughout, I allow for an unobserved determinant of the outcome, ξ_m , that may be correlated with the centralization index. In this case, OLS estimates confound the causal effect of centralization with the effect of the unobserved determinant. I address this issue using the IV strategy described below.

3.1.1 MSA Population and Employment Growth

The MSA population and employment growth equations are similar. Let Y_{mt} denote either population or employment in MSA m at time t , and let T denote the post period of 2017. The specifications come from a multiplicative growth process given by

$$Y_{m,T} = Y_{m,t}^\beta \cdot \exp(\epsilon_{m,t})$$

Where β is a parameter that governs how growth scales with either base population or employment and ϵ is an MSA specific growth shock. Taking the logarithm and moving one unit of $Y_{m,t}$ to the left hand side yields

$$\log \frac{Y_{m,T}}{Y_{m,t}} = (\beta - 1) \log Y_{m,t} + \epsilon_{m,t}$$

To create the empirical specification that I take to the data, I decompose the error term into its mean, the effect of centralization, the effect of time invariant covariates X_m , Census Division fixed effects λ_m , the unobserved determinant of the outcome, and a mean zero idiosyncratic error term as

$$\epsilon_{m,t} = \alpha + \gamma C_m + \delta X_m + \lambda_m + \xi_m + u_{m,t}$$

Substituting in the error term decomposition, I arrive at

$$\log \frac{Y_{m,T}}{Y_{m,t}} = \alpha + (\beta - 1) \log Y_{m,t} + \gamma C_m + \delta X_m + \lambda_m + \xi_m + u_{m,t}$$

I estimate four equations for population and employment growth where only the time index t is varied.

3.1.2 Local Government Employment

Motivated by Figure 2.2, I model local government employment primarily as a function of population, with centralization explaining residual variation conditional on population. I estimate five specifications, one for each survey year and a pooled specification. Let $Y_{m,t}$ denote total local government employment and $P_{m,t}$ denote population in MSA m at time t . The equation I take to the data is

$$\log Y_{m,t} = \alpha + \gamma C_{m,t} + \beta \log P_{m,t} + \delta X_m + \lambda_m + \xi_m + u_{m,t}$$

This estimating equation is similar to the growth equations. However, because I am able to measure a static outcome, local government employment in survey year t , I can use the survey year specific centralization index. I estimate this equation for each of the four ASPEP survey years and a pooled specification that includes survey year specific fixed effects.

3.1.3 Local Government Finance

The specifications I use to estimate the effect of centralization on total revenue and tax revenue largely mirror the specifications for local government employment. Aside from the change in dependent variables, the only alteration is the inclusion of the log ratio of income and employment. Denoting either total revenue or tax revenue in survey year t as Y_{mt} and the ratio of income to employment as W_{mt} , the estimating equation is

$$\log Y_{m,t} = \alpha + \gamma C_{m,t} + \beta \log P_{m,t} + \phi \log W_{m,t} + \delta X_m + \lambda_m + \xi_m + u_{m,t}$$

3.1.4 Data Center Locations

The final empirical specification concerns the location of data centers across MSAs. Because data center locations are measured contemporaneously rather than as a panel, they are not indexed by time. Let $D_m = 1$ if there is a data center in MSA m and zero otherwise. I estimate a linear probability model of the form

$$D_m = \alpha + \gamma C_m + \log P_m + \delta X_m + \lambda_m + \xi_m + u_{m,t}$$

3.2 Rural Free Delivery Instrument

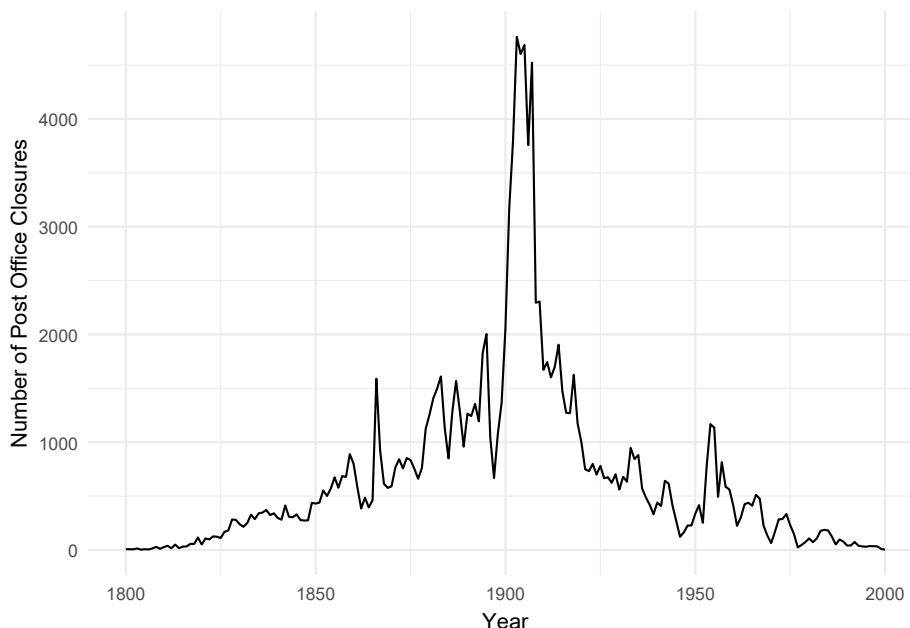
The previous specifications cannot be consistently estimated through OLS if the organization of local government is correlated with unobserved determinants of the outcomes. To address this issue, I employ an instrumental variables strategy based on the Postal Service's implementation of Rural Free Delivery (RFD) in 1902. Using the complete history of U.S. post offices, I instrument centralization with the log number of post office closures within each MSA between 1902 and 1910, the period immediately following the implementation of RFD.

The identifying variation exploited by the instrument comes from differences in the historical dispersion of settlement across MSAs, as measured by post office closures. Because RFD extended the 1863 Free City Delivery program to rural areas, the post offices that closed following its implementation were disproportionately located in what were rural communities in the early twentieth century. Prior to RFD, residents in these communities typically collected mail at local post offices whose viability depended on the surrounding level of postal activity. By shifting delivery away from local pickup and toward direct household

service, RFD reduced the need for many small and geographically dispersed post offices. Variation in the number of closures across MSAs therefore proxies for differences in early settlement dispersion.

Figure 3.1 illustrates the effect of RFD on post office closures. The vertical axis reports the number of post office closures, while the horizontal axis reports the year. The figure displays a large spike in closures during the period immediately following the implementation of RFD.

Figure 3.1: Post Office Closures, 1800-2000



Notes: The figure plots the number of U.S. post office closures by year between 1800 and 2000 using data from @blevins2021. The instrument used in the analysis is the log number of post office closures within an MSA between 1902 and 1910.

I interpret the instrument as affecting modern local government organization through the municipal annexation process. A greater number of historically dispersed settlements increases the number of potential annexation margins within an MSA. Because incorporation is frequently used as a legal defense against annexation, a larger number of potential annexation opportunities increases the likelihood that defensive incorporation occurs somewhere within the metropolitan area, producing a more decentralized local government structure.

Table 3.1 reports the first stage relationship between the RFD instrument and the centralization index. Columns (1)-(4) estimate the first stage separately for each survey year specific centralization index, while column (5) reports the pooled specification with survey year fixed effects included.

Table 3.1: First Stage Estimates

	Centralization Index				
	2017	2007	1997	1987	Pooled
	(1)	(2)	(3)	(4)	(5)
RFD: Number of Post Office Closures	−0.653*** (0.093)	−0.651*** (0.095)	−0.570*** (0.100)	−0.587*** (0.101)	−0.615*** (0.048)
F-Statistic	49.46	50.17	31.54	36.09	168.03
Observations	378	378	378	378	1,512
R ²	0.390	0.364	0.308	0.312	0.338

*p<0.1; **p<0.05; ***p<0.01

Notes: Each column reports a first-stage regression of the centralization index on the log number of post office closures between 1902 and 1910. Columns (1)-(4) use the year-specific index and column (5) reports the pooled specification. All specifications follow the first-stage controls described in the text. Standard errors are reported in parentheses.

The identifying assumption is that historical settlement dispersion, measured by post office closures, affects the modern outcomes I study only through its effect on the organization of local government. I address two related concerns about the exclusion restriction. The first, discussed in the introduction, is that the decisions to accept or resist annexation underlying the effect of the instrument may depend on the same determinants generating the modern endogeneity problem. However, the technologies affecting economies of scale and preferences over public goods are unlikely to remain stable over century-long horizons. Technologies governing transportation, communication, and public service provision change substantially over time, while preferences over local public goods also evolve. Decisions affecting the organization of local government in earlier periods may therefore reflect the determinants of that time without corresponding closely to their modern counterparts. In contrast, political boundaries established through annexation and defensive incorporation are durable and costly to reorganize.

A second concern is that historical settlement patterns may directly affect modern outcomes through persistent private spatial economic organization. I present two arguments against this objection. First, private economic activity is highly dynamic, with firms adjusting through entry, exit, migration, and relocation. Sustaining a direct channel from historical settlement dispersion to modern outcomes through private spatial organization for more than a century would therefore require substantial persistent frictions. To the extent that such frictions exist, the most plausible sources implicate local government organization itself, making it the more plausible persistence mechanism rather than an independent channel. Second, even if historical settlement dispersion generated persistent polycentric economic organization, such persistence would only violate the exclusion restriction if polycentricity

itself exerted an independent effect on the outcomes I study. However, the literature examining the relationship between polycentric metropolitan structure and economic growth remains inconclusive (Li and Liu (2022)). Taken together, these considerations suggest that any long-run persistence arising from historical settlement patterns is more plausibly mediated through the durable organization of local government than through persistent private spatial organization alone.

3.2.1 First Stage Robustness

In this subsection, I report three robustness checks for the first stage. I examine heterogeneity across population size and geography, and assess the linearity and monotonicity assumptions. To examine heterogeneity by population size, I interact the instrument with an indicator for whether an MSA’s population is above the median in the corresponding survey year. Table 3.2 reports the results.

Table 3.2: First Stage Estimates, Population Heterogeneity

	Centralization Index			
	2017	2007	1997	1987
	(1)	(2)	(3)	(4)
RFD: Number of Closures $\times$ Low Population	−0.672*** (0.101)	−0.652*** (0.103)	−0.590*** (0.109)	−0.593*** (0.110)
RFD: Number of Closures $\times$ High Population	−0.533*** (0.096)	−0.538*** (0.098)	−0.424*** (0.103)	−0.424*** (0.104)
Observations	378	378	378	378
R ²	0.351	0.321	0.264	0.261

*p<0.1; **p<0.05; ***p<0.01

Notes: Each column reports a first-stage regression that interacts the log number of post office closures between 1902 and 1910 with an indicator for whether MSA population in that year is above the within-year median. The dependent variable is the year-specific centralization index. Aside from the interaction terms shown, the specifications follow the first-stage controls described in the text. Standard errors are reported in parentheses.

The table shows that the instrument predicts centralization in both large and small MSAs. The estimated effect is somewhat larger among smaller-population MSAs, although the difference is not statistically significant. This pattern suggests that the relationship between historical settlement dispersion and modern local government organization is not driven exclusively by either large or small metropolitan areas.

To address heterogeneity by geography, I follow a similar strategy and interact the instrument with Census Division fixed effects in Table 3.3.

Table 3.3: First Stage Estimates, Geographic Heterogeneity

	Centralization Index			
	2017	2007	1997	1987
	(1)	(2)	(3)	(4)
RFD: Number of Closures x New England	-1.334*** (0.177)	-1.340*** (0.179)	-1.225*** (0.188)	-1.213*** (0.188)
RFD: Number of Closures x Middle Atlantic	-0.725*** (0.112)	-0.696*** (0.114)	-0.644*** (0.119)	-0.727*** (0.121)
RFD: Number of Closures x South Atlantic	-0.523*** (0.099)	-0.539*** (0.101)	-0.506*** (0.106)	-0.502*** (0.106)
RFD: Number of Closures x East South Central	-0.595*** (0.100)	-0.604*** (0.102)	-0.522*** (0.106)	-0.505*** (0.107)
RFD: Number of Closures x West South Central	-0.671*** (0.114)	-0.663*** (0.116)	-0.539*** (0.122)	-0.567*** (0.123)
RFD: Number of Closures x East North Central	-0.855*** (0.100)	-0.854*** (0.101)	-0.794*** (0.107)	-0.817*** (0.109)
RFD: Number of Closures x West North Central	-0.871*** (0.111)	-0.889*** (0.112)	-0.796*** (0.118)	-0.867*** (0.119)
RFD: Number of Closures x Mountain	-0.749*** (0.169)	-0.689*** (0.171)	-0.633*** (0.178)	-0.660*** (0.179)
RFD: Number of Closures x Pacific	-0.667*** (0.135)	-0.652*** (0.137)	-0.633*** (0.143)	-0.690*** (0.144)
Observations	378	378	378	378
R ²	0.346	0.320	0.266	0.270

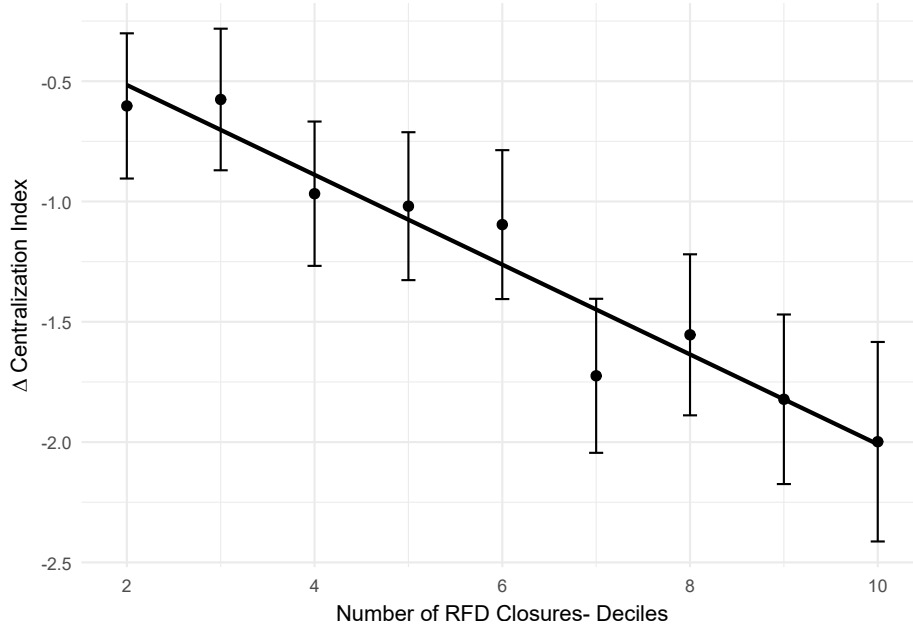
*p<0.1; **p<0.05; ***p<0.01

Notes: Each column reports a first-stage regression that allows the effect of the log number of post office closures between 1902 and 1910 to vary by Census division. The dependent variable is the year-specific centralization index. Aside from the interaction terms shown, the specifications follow the first-stage controls described in the text. Standard errors are reported in parentheses.

Table 3.3 shows that the first-stage relationship is broadly similar across Census divisions. The largest deviation from the average effect appears in New England, although the region contains only 15 MSAs, or roughly 4% of the sample. Overall, these results suggest that the relationship between historical settlement dispersion and modern local government organization is not driven by a particular geographic region.

The final robustness check examines the assumptions of linearity and monotonicity. Instead of modeling the number of post office closures as a continuous instrument, I divide the instrument into deciles and estimate the first stage using the resulting bins. Figure 3.2 plots the estimated decile effects on centralization along with 95% confidence intervals.

Figure 3.2: First Stage Estimates, Effect of Instrument Deciles



Notes: The figure plots coefficients from a pooled first-stage regression that replaces the continuous instrument with deciles of the number of post office closures. Points report coefficient estimates for deciles 2 through 10 relative to the first decile, vertical bars report 95% confidence intervals, and the solid line is a linear fit through the estimated effects.

The figure shows that the relationship between the number of closures and centralization is well approximated by a linear specification. While the estimated effects are not perfectly monotone across deciles, the relationship is generally decreasing in the number of closures, with no evidence of substantial sign reversals or discontinuities. This pattern is consistent with an approximately monotone first-stage relationship and suggests that the identifying variation is not driven by a narrow subset of MSAs or by extreme values of the instrument.

4 Results

In this section, I present the IV estimates for the effect of centralization. I begin by estimating the effect of centralization on population and employment growth. I then examine its effects on local government employment, total revenue, and tax revenue. I conclude the empirical section by estimating the effect of centralization on data center siting decisions.

The tables report only the estimated effect of centralization on each outcome. Each regression corresponds to the specification outlined in Section 3 and includes log population, log land and surface water area, time-invariant covariates, and Census division fixed effects. The

pooled specifications additionally include survey year fixed effects. Because the construction of the index removes natural units from the measure of centralization, the estimates identify only the sign and relative magnitudes of the underlying relationships. The index is standardized and oriented so that the coefficients correspond to a one standard deviation increase in centralization.

Each table also reports two diagnostic statistics. First, the Kleibergen-Paap rk Wald F-statistic evaluates the relevance and strength of the excluded instrument while remaining robust to heteroskedasticity. Larger values indicate that the instrument more strongly predicts centralization (Kleibergen and Paap (2006)).

The second diagnostic is the Anderson-Rubin p-value (Anderson and Rubin (1949); Andrews, Stock, and Sun (2019)), which provides weak-instrument-robust inference by testing whether a candidate parameter value is consistent with the reduced-form relationship between the instrument and outcome. The full procedure constructs confidence sets by testing a sequence of candidate parameter values. The p-value reported in the tables corresponds to the test in which the candidate value is zero. A small value indicates that zero would not be included in the confidence set even without relying on a strong first stage. Together, these diagnostics assess both instrument strength and whether the estimated effects remain distinguishable from zero under inference procedures that remain valid under weak instruments.

I first present the estimated effect of centralization on population and employment growth in Table 4.1. Panel A reports the effects of centralization on population growth. Columns (1)-(4) report the effect of centralization for different base years with column (1) reporting the shortest time horizon, 2007 to 2017, and column (4) reporting the longest horizon, 1977 to 2017. Table 4.1 Panel B follows the same pattern with employment growth as the dependent variable.

Table 4.1: Centralization on Population and Employment Growth

	Growth From Base Year to 2017			
	Base: 2007	Base: 1997	Base: 1987	Base: 1977
	(1)	(2)	(3)	(4)
<i>Panel A: Population Growth</i>				
Centralization Index: 2007	0.009 (0.007)			
Centralization Index: 1997		0.037* (0.019)		
Centralization Index: 1987			0.067** (0.030)	0.122*** (0.041)
KP rk Wald F-statistic	48.05	30.21	34.56	35.53
Anderson–Rubin p-value	0.1551	0.035	0.0155	0.0012
Observations	378	378	378	378
R ²	0.323	0.274	0.295	0.349
<i>Panel B: Employment Growth</i>				
Centralization Index: 2007	0.020** (0.010)			
Centralization Index: 1997		0.070*** (0.023)		
Centralization Index: 1987			0.056* (0.029)	0.097** (0.041)
KP rk Wald F-statistic	48.74	31.59	36.95	36.98
Anderson–Rubin p-value	0.025	6e-04	0.0432	0.0155
Observations	378	378	378	378
R ²	0.063	0.127	0.341	0.368

*p<0.1; **p<0.05; ***p<0.01

Notes: Two-stage least squares estimates of the effect of centralization on metropolitan population growth (Panel A) and private employment growth (Panel B). Columns differ by the base year used to construct growth through 2017. Each regression follows the growth specification described in the text. Kleibergen-Paap rk Wald F-statistics and Anderson-Rubin p-values are reported below the coefficient estimates. Standard errors are reported in parentheses.

Table 4.1 reports positive coefficients for centralization in both panels. In Panel A, the population growth coefficients are positive in all four specifications, although the 2007 to 2017 estimate is small and not statistically distinguishable from zero. In Panel B, all four employment growth coefficients are positive and statistically significant. Across both measures, the positive relationship between centralization and growth is more consistent with coordination models than with Tiebout-style models of competition. The sign of the estimated effect corresponds to the key comparative static over which these frameworks differ: mod-

els of Tiebout competition predict that greater decentralization increases growth through interjurisdictional competition, while models emphasizing coordination predict that greater centralization increases growth by internalizing externalities.

I next examine local government employment as a measure of public input use. Table 4.2 reports four cross-sectional specifications and one pooled specification for the effect of centralization on local government employment. Column (1) reports the estimate using the 2017 ASPEP data. Columns (2)-(4) report the corresponding estimates for 2007, 1997, and 1987 respectively. Column (5) reports the pooled specification using the centralization index constructed from the joint variation over time.

Table 4.2: Centralization on Local Government Employment

	Local Government Employment				
	2017	2007	1997	1987	Pooled
	(1)	(2)	(3)	(4)	(5)
Centralization Index: 2017	0.054** (0.027)				
Centralization Index: 2007		0.077*** (0.027)			
Centralization Index: 1997			0.081** (0.033)		
Centralization Index: 1987				0.107*** (0.034)	
Centralization Index: Pooled					0.077*** (0.015)
KP rk Wald F-statistic	47.36	48.05	30.21	34.56	165.92
Anderson–Rubin p-value	0.0319	0.0021	0.0087	5e-04	0
Observations	378	378	378	378	1,512
R ²	0.961	0.961	0.955	0.953	0.957

*p<0.1; **p<0.05; ***p<0.01

Notes: Two-stage least squares estimates of the effect of centralization on log local government employment. Columns (1)-(4) report year-specific cross sections and column (5) reports the pooled specification. Each regression follows the employment specification described in the text. Kleibergen-Paap rk Wald F-statistics and Anderson-Rubin p-values are reported below the coefficient estimates. Standard errors are reported in parentheses.

Table 4.2 reports positive coefficients in every specification. All five estimates are statistically significant, and the coefficients are similar in magnitude across columns. The positive relationship between centralization and local government employment suggests that the growth effects documented in Table 4.1 are unlikely to arise from a simple labor-saving economies of scale mechanism where centralized MSA increase expected utility by passing cost savings onto residents.

I next examine local government finances. Table 4.3 contains two panels with an identical structure to Table 4.2. Panel A reports the effect of centralization on total revenue, while Panel B reports the effect for tax revenue.

Table 4.3: Centralization on Total and Tax Local Government Revenue

	<i>Dependent variable:</i>				
	2017	2007	1997	1987	Pooled
	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Total Revenue</i>					
Centralization Index: 2017	0.142*** (0.050)				
Centralization Index: 2007		0.168*** (0.051)			
Centralization Index: 1997			0.142** (0.059)		
Centralization Index: 1987				0.095* (0.051)	
Centralization Index: Pooled					0.135*** (0.026)
KP rk Wald F-statistic	38.05	35.53	22.68	32.88	137.68
Anderson–Rubin p-value	0.002	2e-04	0.0079	0.0598	0
Observations	378	378	378	378	1,512
R ²	0.907	0.913	0.913	0.914	0.925
<i>Panel B: Tax Revenue</i>					
Centralization Index: 2017	0.107** (0.044)				
Centralization Index: 2007		0.151*** (0.046)			
Centralization Index: 1997			0.098* (0.056)		
Centralization Index: 1987				0.139** (0.055)	
Centralization Index: Pooled					0.123*** (0.025)
KP rk Wald F-statistic	38.05	35.53	22.68	32.88	137.68
Anderson–Rubin p-value	0.0091	3e-04	0.0651	0.0073	0
Observations	378	378	378	378	1,512
R ²	0.931	0.933	0.927	0.912	0.937

*p<0.1; **p<0.05; ***p<0.01

Notes: Two-stage least squares estimates of the effect of centralization on log total revenue (Panel A) and log tax revenue (Panel B). Columns (1)-(4) report year-specific cross sections and column (5) reports the pooled specification. Each regression follows the finance specification described in the text. Kleibergen-Paap rk Wald F-statistics and Anderson-Rubin p-values are reported below the coefficient estimates. Standard errors are reported in parentheses.

Table 4.3 reports positive coefficients for both total revenue and tax revenue in every specification. In both panels, the estimates are statistically distinguishable from zero at conventional levels and are similar in magnitude across specifications. The positive relationship between centralization and tax revenue is consistent with models in which decentralized governments underprovide public goods because they fail to internalize the effects of taxation on neighboring jurisdictions' tax bases.

Taken together, Tables 4.2 and 4.3 suggest that the positive growth effects associated with centralization are unlikely to arise from a simple labor-saving economies of scale mechanism. More centralized MSAs employ more local government labor and collect more revenue conditional on population, a pattern more consistent with centralization relaxing coordination constraints on public good provision than with reducing input use through scale economies.

I conclude by examining data center locations, which provide a close empirical analogue of mobile capital in the tax competition framework. Table 4.4 reports the corresponding estimate.

Table 4.4: Centralization on Data Center Location Decisions

	<i>Dependent variable:</i>
	Data Center in MSA
Centralization Index	−0.147** (0.072)
KP rk Wald F-statistic	43.63
Anderson–Rubin p-value	0.0281
Observations	378
R ²	0.285

*p<0.1; **p<0.05; ***p<0.01

Notes: Two-stage least squares estimate of the effect of centralization on whether an MSA hosts at least one data center facility. The dependent variable is an indicator equal to one if the MSA contains at least one data center. The regression follows the data center specification described in the text. Kleibergen–Paap rk Wald F-statistic and Anderson–Rubin p-value are reported below the coefficient estimate. Standard errors are reported in parentheses.

Table 4.4 reports a negative and statistically significant coefficient for centralization in the data center specification. The estimate suggests that decentralized MSAs compete more aggressively for highly mobile investment, consistent with the predictions of tax competition models. More broadly, the result suggests that coordination failures associated with fragmented local government organization are sufficiently strong to affect firm location decisions.

Taken together, the results suggest that the organization of local government plays an important role in shaping metropolitan economic outcomes. More centralized MSAs exhibit

higher levels of population and employment growth, greater local government employment, and higher revenue collection conditional on population. At the same time, decentralized MSAs appear more likely to attract highly mobile investment in the form of data centers. Viewed jointly, these patterns suggest that decentralization may intensify competition for mobile capital while limiting the ability of governments to coordinate taxation and public good provision at the metropolitan level. Overall, the results are more consistent with models emphasizing coordination failures under fragmented local government organization than with models in which decentralization primarily improves outcomes through Tiebout-style competition.

5 Conclusion

This paper studies the effect of local government organization on metropolitan growth. I measure centralization across U.S. metropolitan statistical areas using the distribution of employment across local governments and estimate its causal effect using an instrument based on post office closures induced by the introduction of Rural Free Delivery in 1902. The instrument captures historical settlement dispersion, which I argue shaped modern local government fragmentation through the annexation and defensive incorporation process. Using this strategy, I find that more centralized MSAs experience higher population and employment growth. I then examine local government employment, total revenue, tax revenue, and data center locations to distinguish among competing mechanisms. The results show that centralized MSAs employ more local government labor and collect more revenue, while decentralized MSAs are more likely to attract data centers, a form of highly mobile, tax-sensitive capital.

Metropolitan areas act as integrated economic units, but their political authority is often fragmented across many jurisdictions. Local government organization matters because local governments do not act in isolation, and changes in organization change the strategic environment in which local governments operate. The evidence presented in this paper suggests that an important margin in local government interactions is coordination failure. When each local government makes decisions over only the costs and benefits within its own jurisdiction, competition over the same mobile residents, firms, and fiscal base can leave cross-jurisdictional externalities unaccounted for, reducing aggregate fiscal capacity and limiting the provision of public goods whose benefits extend across jurisdictional boundaries. In this setting, centralization can improve outcomes by moving decision-making closer to the scale of the metropolitan economy and internalizing spillovers that decentralized governments do

not fully take into account.

The mechanism evidence strengthens this interpretation. If centralization raised growth mainly through labor-saving economies of scale, we would expect centralized MSAs to use fewer public inputs conditional on population. Instead, they employ more local government labor and collect more revenue. This pattern suggests that centralization is not simply reducing the cost of a fixed bundle of public goods. It appears to expand the fiscal and administrative capacity used to provide them. At the same time, decentralized MSAs are more likely to attract data centers. This pattern is consistent with tax competition: fragmented jurisdictions may compete aggressively for mobile capital while also generating lower aggregate revenue and weaker growth at the metropolitan level.

Centralization need not have the same effects at every scale of government. At broader geographic scales, the costs of centralization may be substantial: preferences are more heterogeneous, local information is harder to aggregate, accountability may weaken, and spillovers across jurisdictions may be less dense. In that setting, decentralization can improve outcomes by allowing policy to vary across places and by preserving competitive discipline among governments. Metropolitan areas present a different tradeoff. Local governments within an MSA are geographically close and economically linked through shared labor markets, infrastructure networks, mobile residents, firms, and overlapping tax bases. Because these interactions occur within a tightly connected spatial economy, the spillovers from local policy choices are likely to be more salient. The results of this paper suggest that, at the metropolitan scale, centralization can improve outcomes when the benefits of internalizing these spillovers exceed the costs of reducing interjurisdictional competition.

This paper's focus on metropolitan level growth is useful because growth summarizes many dimensions of local government performance in a single outcome. But that aggregation also limits what the estimate can show. The effect of centralization on growth is not a single mechanism; it reflects the combined effects of centralization across many margins, including public safety, schools, infrastructure, land use, permitting, taxation, environmental regulation, and the reallocation of resources across space. The aggregate estimate therefore helps identify which framework of local government behavior is most consistent with the broad pattern of outcomes, but it should not be taken to imply that coordination failures are always the driving factor or that all local government provision is more efficient under centralization. A natural next step is to move below the aggregate growth estimate and examine the particular public goods and policy margins through which local government organization shapes economic activity.

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